

## Why use $10^9 + 7$ ?

In competitive programming, we often deal with **very large numbers**. To prevent overflow and keep numbers manageable, we calculate answers **modulo**  $10^9 + 7$ .

## Basic Modulo Operation

►Syntax in C++: `const int MOD = 1e9 + 7;`

Use this constant when doing operations like addition, subtraction, multiplication.

## Addition Modulo

```
int a = 1000000000;  
int b = 50;  
int result = (a + b) % MOD;
```

Always use `% MOD` after the operation.